Paper structure

hybrid

Questions → Netexam

50%

50% = 100

25 MCQ

Labshee Lecture Workshop

drag and drop
drop down selection
3

Essay

25 marks ① Supervise / Unsupervised

25 marks ② DNN

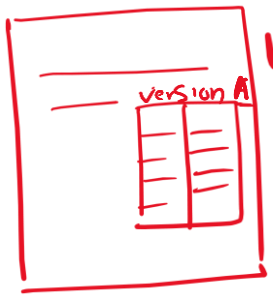
② Versions

① ②

25 MCQ

Essay Version A ①

booklet



Comprehensive Sample Question

Scenario

A startup named**Health AI** is developing an **AI-based smart health risk monitoring platform** that predicts health risk based on environmental and physiological data collected via sensors and wearables.
You are part of the AI engineering team responsible for developing, evaluating, and ensuring the ethical use of the system.

(a) Neuron Computation (5 marks)

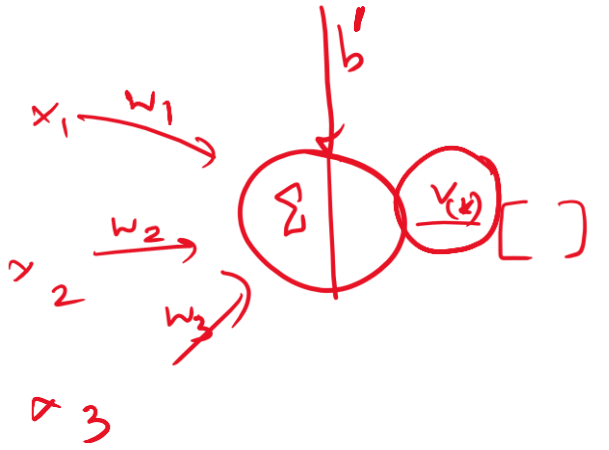
The output layer of the ANN receives the following inputs for health risk prediction:

Input	Description	Value	Weight
x_1	Normalized heart-rate deviation	0.6	0.5
x_2	Sleep-quality index	0.3	0.4
x_3	Daily stress level	0.8	−0.3

Bias $b = 0.2$
Activation function $\phi(v) = sgn(x)$

Compute:

1. The net input v_k → Show all the steps
2. The neuron output y_k



$v_k = x_1 w_1 + x_2 w_2 + x_3 w_3 + b$

$v_k = 0.6 \times 0.5 + 0.3 \cdot 0.4 + 0.8 \times -0.3 + 0.2$

Compute Net Input v_k

$$v_k = w_1x_1 + w_2x_2 + w_3x_3 + b$$

Substitute the given values:

$$v_k = (0.5)(0.6) + (0.4)(0.3) + (-0.3)(0.8) + 0.2$$

Compute each term:

1. $(0.5)(0.6) = 0.30$
2. $(0.4)(0.3) = 0.12$
3. $(-0.3)(0.8) = -0.24$
4. Bias $= +0.20$

Add them up:

$$v_k = 0.30 + 0.12 - 0.24 + 0.20 = 0.38$$

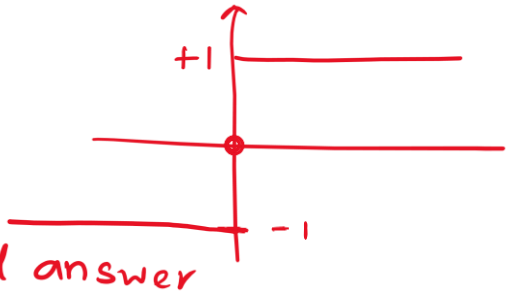
→ include all the steps to get full marks

Compute Neuron Output y_k

Since $v_k = 0.38 > 0$,

$$y_k = \text{sgn}(0.38) = +1$$

Predicted answer



Neuron output $y_k = +1$

(b) The neuron from part (a) predicted $+1$for health risk, but the **true label** is -1. Using a learning rate of 0.05, perform **one** Perceptron update to **adjust the three weights and the bias**. Show the misclassification check, compute the scalar update factor, and provide all intermediate calculations leading to the new values **of w_1, w_2, w_3,and b.**

(6 marks)

Given

- Inputs $x = [x_1, x_2, x_3] = [0.6, 0.3, 0.8]$
- Current weights $w = [w_1, w_2, w_3] = [0.5, 0.4, -0.3]$
- Current bias $b = 0.2$
- From part (a): predicted $y = +1$
- Target $t = -1$
- Learning rate $\eta = 0.05$

equation $W_{new} = W + \eta (actual - Predicted) x$

$W_1^{new} = 0.5 + 0.05 (-1 - (+1)) 0.6$

$= 0.4$

$W_2^{new} = W_2 + \eta [a - p] x_2$

$= 0.4 + 0.05 [-1 - (+1)] 0.3$

$= 0.4$

$W_3^{new} = W_3 + \eta [a - p] x_3$

$= -0.3 + 0.05 [-1 - [+1]) 0.8$

$= -0.3$

$b_n^{new} = b + \eta [a - p] 1$

$= 0.2$

Perceptron update rule (bipolar targets, using $(t - y)$form)

$$w' = w + \eta (t - y) x, \quad b' = b + \eta (t - y)$$

w_2

w_3

$b'_{new} = b + \eta (a - p) 1$

$= 0.2$

1. **Compute the error factor**

$$t - y = (-1) - (+1) = -2$$
$$\eta\,(t - y) = 0.05 \times (-2) = -0.10$$

2. **Weight increments**

$$\Delta w_1 = (-0.10) \times 0.6 = -0.06, \Delta w_2 = (-0.10) \times 0.3 = -0.03, \Delta w_3 = (-0.10) \times 0.8 = -0.08$$

3. **Updated weights**

$$w_1' = 0.5 + (-0.06) = 0.44, w_2' = 0.4 + (-0.03) = 0.37, w_3' = -0.3 + (-0.08) = -0.38$$

4. **Bias update**

$$\Delta b = -0.10, b' = 0.2 + (-0.10) = 0.10$$

Final result

- **Updated weights:** $[0.44,\ 0.37,\ -0.38]$
- **Updated bias:** 0.10

(c) Model Behavior Analysis (4 marks)

During model training for **HEALTH risk prediction**, the following results were observed:

Epoch	Training Accuracy	Validation Accuracy	Training Loss	Validation Loss
10	85%	83%	0.42	0.45
20	93%	86%	0.22	0.33
30	97%	81%	0.14	0.52
40	99%	75%	0.08	0.70

i. Identify the problem illustrated by the table. (what can you tell about model performance)
ii. Suggest **two machine learning techniques** that can be applied to improve the model's generalization performance, and briefly explain how each helps.

explain each technique

(d) Model Selection and Justification (5 × 2 = 10 marks)

Select the most appropriate **deep learning model (FNN, CNN, LSTM, GAN)** for each of the following tasks and **justify briefly**:

i. Predicting farmer health risk from tabular health records (age, BMI, glucose level, sleep hours).

ii. Detecting crop diseases from leaf images captured via drone.

iii. Forecasting next week's soil-moisture level using 30 days of sensor data. - LSTM

iv. Generating realistic new crop images for data augmentation.- GAN

v. Predicting credit default likelihood from customer financial history. - FNN

you may why other models are not suitable

→ gender bias

(e) Ethics and Bias (4 marks)

After deployment, the health-risk model performs significantly better for **male users** than **female users** even with similar lifestyle and medical data.

i. Identify **two possible types of bias** present in this scenario.

ii. how this biasness have occur?

ii. Suggest **one technical** and **one organizational** strategy to mitigate these biases and promote ethical fairness.